

Lesson 3-1: Understand Ratios

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute 8×7 .

Answer: 56

Use $8 \times (5 + 2) = 40 + 16$.

Prerequisite skill: Recall multiplication facts through 12×12

TASK 2 • FOUNDATION 2

Continue the pattern: 6, 12, 18, ____.

Answer: 24

Each step adds 6.

Prerequisite skill: Skip-count by any one-digit number

TASK 3 • FOUNDATION 3

There are 5 blue counters and 7 red counters.
How many counters in all?

Answer: 12

Add both parts: $5 + 7 = 12$.

Prerequisite skill: Count each part of a part-whole picture, then add the parts for the total

TASK 4 • FOUNDATION 4

Three of eight equal slices are shaded. Write the shaded fraction.

Answer: $\frac{3}{8}$

The numerator counts shaded slices; the denominator counts all equal slices.

Prerequisite skill: Write a fraction naming a part of a whole

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TASK 5 • QUICK CHECK A

6×4 ? There are 3 dogs and 5 cats — how many animals in all?

Answer: $6 \times 4 = 24$. Total animals: 3 dogs + 5 cats = 8 animals total. (Part-to-part ratio is 3 : 5; part-to-whole ratio of dogs is 3 : 8).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

7×3 ? There are 4 red marbles and 6 blue — how many marbles in all?

Answer: $7 \times 3 = 21$. Total: 4 + 6 = 10 marbles. Part-to-part is 4 : 6; red to total is 4 : 10.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

8×5 ? A team has 5 guards and 7 forwards — how many players in all?

Answer: $8 \times 5 = 40$. Total: 5 + 7 = 12 players. Guards to forwards is 5 : 7; guards to total is 5 : 12.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

All 28 students in a class wear either sneakers or boots. The ratio wearing sneakers to boots is 5 : 2. How many wear sneakers?

Answer: $5 + 2 = 7$ parts, and $28 \div 7 = 4$ students per part, so sneakers = $5 \times 4 = 20$.

Check the result against the situation and explain your reasoning.

Lesson 3-2: Rates and Unit Rates

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.2

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $168 \div 12$.

Answer: 14

$12 \times 10 = 120$ and $12 \times 4 = 48$; together they make 168.

Prerequisite skill: Divide a multi-digit number by a one- or two-digit number

TASK 2 • FOUNDATION 2

Five notebooks cost \$15. What is the cost in dollars for one notebook?

Answer: 3

Divide cost by the number of notebooks: $15 \div 5 = 3$ dollars per notebook.

Prerequisite skill: Divide to find the amount for one, and read “per” as “for each one”

TASK 3 • FOUNDATION 3

A table gives 1 bag $\rightarrow$ 4 marbles, 2 bags $\rightarrow$ 8, 3 bags $\rightarrow$ 12. How many marbles for 5 bags?

Answer: 20

Each bag adds 4 marbles; $5 \times 4 = 20$.

Prerequisite skill: Keep a repeating pattern going in a table

TASK 4 • FOUNDATION 4

A cyclist rides 24 miles in 2 hours. State the unit rate with its unit.

Answer: 12 miles per hour

Divide miles by hours: $24 \div 2 = 12$ miles per hour.

Prerequisite skill: Attach the correct unit label to a computed number

Lesson 3-2: Rates and Unit Rates

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.2

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$60 \div 4$? If 4 pencils cost \$2, what does 1 pencil cost?

Answer: $60 \div 4 = 15$. Unit price: 1 pencil costs $\$2.00 \div 4 = \0.50 (50 cents).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$72 \div 6$? If 6 notebooks cost \$9, what does 1 notebook cost?

Answer: $72 \div 6 = 12$. Unit price: $\$9 \div 6 = \1.50 per notebook.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$150 \div 5$? If a car travels 150 miles in 5 hours, what is the speed per hour?

Answer: $150 \div 5 = 30$. Speed: 30 miles per hour.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Brand A is 12 oz for \$3.00. Brand B is 20 oz for \$4.60. Which is the better buy?

Answer: A: $\$3.00 \div 12 = \0.25 per oz. B: $\$4.60 \div 20 = \0.23 per oz. Brand B is the better buy.

Check the result against the situation and explain your reasoning.

Lesson 3-3: Equivalent Ratios Using Tables

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.a

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Scale the pair 3 : 5 by a factor of 4.

Answer: 12 : 20

Multiply both terms by 4: $3 \times 4 = 12$ and $5 \times 4 = 20$.

Prerequisite skill: Multiply both numbers of a pair by the same factor

TASK 2 • FOUNDATION 2

Compute 9×8 .

Answer: 72

$10 \times 8 - 8 = 80 - 8 = 72$.

Prerequisite skill: Recall multiplication facts through 12×12

TASK 3 • FOUNDATION 3

Fill in the numerator: $2/3 = \underline{\quad}/15$.

Answer: 10

The denominator is multiplied by 5, so multiply the numerator by 5 too.

Prerequisite skill: Write equivalent fractions by multiplying numerator and denominator

TASK 4 • FOUNDATION 4

A table lists (2, 6), (4, 12), (6, 18). What pair comes next if the same step continues?

Answer: (8, 24)

Add 2 to the first coordinate and 6 to the second.

Prerequisite skill: Fill the next row of a growing pattern table

Lesson 3-3: Equivalent Ratios Using Tables

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.a

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

2×5 and 3×5 ? Complete: $2 : 3 = 8 : \underline{\hspace{1cm}}$

Answer: $2 \times 5 = 10$ and $3 \times 5 = 15$. Complete ratio: Since $2 \times 4 = 8$, scale 3 by 4: $3 \times 4 = 12$ ($2 : 3 = 8 : 12$).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

4×6 and 5×6 ? Complete: $4 : 5 = 24 : \underline{\hspace{1cm}}$

Answer: $4 \times 6 = 24$ and $5 \times 6 = 30$. Since $4 \times 6 = 24$, scale 5 by 6: $4 : 5 = 24 : 30$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

3×7 and 8×7 ? Complete: $3 : 8 = 21 : \underline{\hspace{1cm}}$

Answer: $3 \times 7 = 21$ and $8 \times 7 = 56$. So $3 : 8 = 21 : 56$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A recipe uses 3 cups flour to 2 cups water. You have 8 cups of water. How much flour?

Answer: Water scaled by 4 ($2 \times 4 = 8$), so flour is $3 \times 4 = 12$ cups.

Check the result against the situation and explain your reasoning.

Lesson 3-4: Equivalent Ratios Using Graphs

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.a

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Describe how to plot (3, 5) from the origin.

Answer: Move 3 units right and 5 units up

The first coordinate is horizontal; the second is vertical.

Prerequisite skill: Plot a point in Quadrant I from an ordered pair

TASK 2 • FOUNDATION 2

The x-axis ticks are 0, 2, 4, 6, 8. What x-value is at the third tick to the right of 0?

Answer: 6

Three intervals of 2 give 6.

Prerequisite skill: Read the scale on each axis before plotting

TASK 3 • FOUNDATION 3

A point is 4 units right and 7 units up from the origin. Give its ordered pair.

Answer: (4, 7)

Write horizontal position first, then vertical position.

Prerequisite skill: Count right from the origin, then up

TASK 4 • FOUNDATION 4

The rule is $y = 3x$. Give y when $x = 4$.

Answer: 12

Substitute 4 for x : $3 \times 4 = 12$.

Prerequisite skill: Build a ratio table to generate the pairs

Lesson 3-4: Equivalent Ratios Using Graphs

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.a

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Start at (0, 0), move 3 units right and 2 units up — write the ordered pair. A horizontal axis has grid lines every 5 units, starting at 0. How many intervals right of 0 is the value 15?

Answer: Ordered pair: (3, 2) — move 3 units right on x-axis, then 2 units up on y-axis. Grid position: If each grid line is 5, 15 is at line 3 ($15 \div 5 = 3$).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Start at (0, 0), move 5 units right and 4 units up — write the ordered pair. A horizontal axis has grid lines every 2 units, starting at 0. How many intervals right of 0 is the value 12?

Answer: Ordered pair (5, 4). With gridlines worth 2, the value 12 sits at line 6 ($12 \div 2 = 6$).

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Start at (0, 0), move 2 units right and 7 units up — write the ordered pair. A horizontal axis has grid lines every 10 units, starting at 0. How many intervals right of 0 is the value 50?

Answer: Ordered pair (2, 7). With gridlines worth 10, the value 50 sits at line 5 ($50 \div 10 = 5$).

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A ratio graph passes through (0, 0), (2, 6), and (4, 12). What is the unit rate, and where is $x = 5$?

Answer: Unit rate = $6 \div 2 = 3$, so $y = 3x$. At $x = 5$, $y = 15$ — the point (5, 15).

Check the result against the situation and explain your reasoning.

Lesson 3-5: Comparing Ratio Relationships

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.b

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Which is greater: 35 or 53?

Answer: 53

Compare tens: 5 tens is greater than 3 tens.

Prerequisite skill: Compare two whole numbers

TASK 2 • FOUNDATION 2

Scale 4 : 7 to an equivalent ratio with first term 20.

Answer: 20 : 35

Multiply both terms by 5.

Prerequisite skill: Scale both parts of a ratio to hit a target value

TASK 3 • FOUNDATION 3

What is the least common multiple of 6 and 8?

Answer: 24

Multiples of 6 are 6, 12, 18, 24; multiples of 8 are 8, 16, 24.

Prerequisite skill: Find a common multiple of two numbers

TASK 4 • FOUNDATION 4

Pack A costs \$12 for 4 pens. Pack B costs \$14 for 7 pens. Which has the lower price per pen?

Answer: Pack B

A costs \$3 per pen; B costs \$2 per pen.

Prerequisite skill: Divide to find a unit rate for a fair comparison

Lesson 3-5: Comparing Ratio Relationships

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.b

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Which is greater, 12 or 9? Scale 3 : 4 so the first number is 12.

Answer: 12 is greater than 9. Scale 3 : 4: Since $3 \times 4 = 12$, multiply 4 by 4 to get 16. Scaled ratio: 12 : 16.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Which is greater, 18 or 15? Scale 2 : 5 so the first number is 18.

Answer: 18 is greater. Since $2 \times 9 = 18$, multiply 5 by 9: $2 : 5 = 18 : 45$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Which is greater, 24 or 30? Scale 4 : 7 so the first number is 24.

Answer: 30 is greater. Since $4 \times 6 = 24$, multiply 7 by 6: $4 : 7 = 24 : 42$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Runner A covers 3 miles in 24 minutes; Runner B covers 5 miles in 45 minutes. Who is faster?

Answer: A: $24 \div 3 = 8$ minutes per mile. B: $45 \div 5 = 9$ minutes per mile. Runner A is faster (fewer minutes per mile).

Check the result against the situation and explain your reasoning.

Lesson 3-6: Converting Within One System

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.d

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $72 \div 9$.

Answer: 8

Check with $9 \times 8 = 72$.

Prerequisite skill: Recall multiplication and division facts

TASK 2 • FOUNDATION 2

Use 1 foot = 12 inches. How many inches are in 3 feet?

Answer: 36

Multiply $3 \times 12 = 36$ inches.

Prerequisite skill: Recall customary facts: 1 ft = 12 in, 1 lb = 16 oz, 1 gal = 4 qt

TASK 3 • FOUNDATION 3

How many meters are in 2.4 kilometers? Use 1 km = 1000 m.

Answer: 2400

Multiply 2.4 by 1000 because each kilometer contains 1000 meters.

Prerequisite skill: Multiply and divide by 10, 100, 1000 for metric steps

TASK 4 • FOUNDATION 4

Convert 48 inches to feet using 12 inches = 1 foot.

Answer: 4

Divide by 12 because 12 inches form one foot.

Prerequisite skill: Decide whether a conversion calls for multiplying or dividing

Lesson 3-6: Converting Within One System

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.d

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

12×4 ? 1 foot = ____ inches? 2.5 m = ____ cm?

Answer: $12 \times 4 = 48$. 1 foot = 12 inches. 2.5 m = 250 cm ($2.5 \times 100 = 250$).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

11×4 ? 1 pound = ____ ounces? 3.5 m = ____ cm?

Answer: $11 \times 4 = 44$. 1 pound = 16 ounces. 3.5 m = 350 cm (3.5×100).

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

9×6 ? 1 gallon = ____ quarts? 0.75 km = ____ m?

Answer: $9 \times 6 = 54$. 1 gallon = 4 quarts. 0.75 km = 750 m (0.75×1000).

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A tank holds 3 gallons. How many cups is that? (1 gal = 4 qt, 1 qt = 4 cups)

Answer: $3 \text{ gal} \times 4 = 12 \text{ qt}$, and $12 \text{ qt} \times 4 = 48 \text{ cups}$.

Check the result against the situation and explain your reasoning.

Lesson 3-7: Converting Between Systems

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.d

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute 6×2.5 .

Answer: 15

$$6 \times 2 + 6 \times 0.5 = 12 + 3.$$

Prerequisite skill: Multiply a whole number by a decimal

TASK 2 • FOUNDATION 2

Round 7.846 to the nearest hundredth.

Answer: 7.85

The thousandths digit is 6, so round the hundredths digit up.

Prerequisite skill: Round a decimal to the nearest tenth or hundredth

TASK 3 • FOUNDATION 3

Use 1 inch = 2.54 centimeters. Convert 5 inches to centimeters.

Answer: 12.7

$$5 \times 2.54 = 12.70 \text{ centimeters.}$$

Prerequisite skill: Multiply by a conversion factor and keep the unit label

TASK 4 • FOUNDATION 4

Use 1 yard = 3 feet and 1 foot = 12 inches. Convert 2 yards to inches.

Answer: 72

First $2 \times 3 = 6$ feet, then $6 \times 12 = 72$ inches.

Prerequisite skill: Chain two conversions in order without losing a step

Lesson 3-7: Converting Between Systems

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.d

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

4×1.6 ? 10×0.45 ? Round 6.44 to the nearest tenth.

Answer: $4 \times 1.6 = 6.4$. $10 \times 0.45 = 4.5$. Round 6.44 to nearest tenth = 6.4 (since the hundredths digit 4 is less than 5).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

5×2.54 ? 8×0.45 ? Round 12.7 to the nearest tenth.

Answer: $5 \times 2.54 = 12.7$. $8 \times 0.45 = 3.6$. 12.7 is already to the nearest tenth: 12.7.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

3×1.6 ? 20×0.45 ? Round 9.08 to the nearest tenth.

Answer: $3 \times 1.6 = 4.8$. $20 \times 0.45 = 9.0$. Rounding 9.08 to the nearest tenth gives 9.1.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A race is 5 kilometers. About how many miles is that, if $1 \text{ km} \approx 0.62 \text{ mi}$? Round to the nearest tenth.

Answer: $5 \times 0.62 = 3.1$ miles.

Check the result against the situation and explain your reasoning.